12 · Expected Impact & Regulation

> **Assignment:** *“Describe expected impact — what is the potential impact (positive and negative) of the CDSS? What benefits and challenges does the group anticipate?"*

12.1 The Guiding Principle

> **For a triage tool in the hands of patients, false reassurance weighs considerably more heavily than a false alarm.**
>
> A false alarm costs a phone call. False reassurance can delay an infection by days. For this reason our logic is consistently designed **worst-first** and deliberately accepts a higher false alarm rate — and for this reason the proportion of GREEN cases that nevertheless present to a physician is our most important safety indicator.

This asymmetry runs through all design decisions: through the order of the rules (06), through the decision not to place safety-critical questions behind conditional logic (B-03), and through the prioritisation of the test cases (10).

12.2 Expected Benefits

Level	Benefit	Mechanism of action
Patient	Earlier treatment in the event of infection	Structured daily data capture detects deterioration that goes unnoticed in everyday life
	Less unnecessary worry and fewer calls	Active all-clear for normal signs of healing (scab, slight redness) — Scenario A
	Orientation instead of uncertainty	Concrete instructions for action instead of “report anything unusual"
Health Worker	Triage in seconds instead of minutes	A structured alert with classification, vital signs and trend replaces reconstructing the situation over the phone
	Fewer contacts without clinical consequence	Normal signs of healing do not reach the practice in the first place
Hospital / practice	Complete trend record for the follow-up examination	Daily data points instead of retrospective patient recollection
	Quality data on the practice's own SSI rate	Aggregated analysis across procedures and surgeons
Health system	Potentially fewer readmissions and revision surgeries	Earlier intervention in superficial SSI prevents progression
	Structured, standardised follow-up care data	FHIR-based, bound to terminology, analysable

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12.3 Expected Harm and Risks

Risk	Mechanism	Mitigation
False reassurance	The patient accepts GREEN even though an infection is present — for instance because the image assignment failed (C-01)	Worst-first logic; ungated safety questions; the note “when in doubt, always call" in every result; monitoring of GREEN cases that present to a physician
Automation bias / deskilling	The patient relies entirely on the app and no longer observes the wound personally	The result always states the **reason** for the classification, not just the colour — this preserves the patient's own observation
Alert fatigue	Too many ORANGE notifications → the practice skims or ignores them	Alerts only from ORANGE upwards; override rate as a monitoring indicator with an alarm threshold
Systematic exclusion	Patients with findings outside the algorithm (scope limits) receive worse care than before because the system suggests safety	Scope note in the app; rule R8 never phrases things as “everything is fine" without the addition that patients should get in touch if worried
Scaling of one's own errors	An incorrect rule is now applied **consistently** incorrectly — to hundreds of patients. A CDSS-specific risk: systematisation amplifies errors instead of averaging them out	Versioning in the FHIR export makes it reconstructible which rule version produced which recommendation; incident process; regression tests with every change
Digital divide	Anyone without a smartphone, without internet or without reading literacy does not benefit — the care gap grows rather than shrinks	Image-based operation lowers the language and reading barrier; surveying the proportion without access as an equity indicator; conventional follow-up care must continue to exist in parallel
Data protection	Health data are transmitted by the patient into the record — consent, transmission security, purpose limitation	Explicit consent; no data collection without a purpose of use (the reason for fixing B-05); encryption
More work instead of relief	An additional channel that has to be serviced — without phone calls decreasing	Consultation duration and contact numbers are measured explicitly; if both increase, the intervention has failed
Expectation of permanent availability	An ORANGE at 23:00 creates the expectation of an immediate response	Communicate the time window for alert processing during the discharge consultation; RED always additionally refers to the emergency service
Amplification of anxiety	Daily confrontation with one's own wound increases worry in some people instead of reducing it	Collect this as an outcome in the pilot evaluation; offer an opt-out option
Liability	Who is responsible for an incorrect recommendation — the manufacturer, the hospital, the treating physician?	Disclaimer; positioning as a triage instrument, not a diagnostic one; regulatory clarification before any deployment
Sustainability	What happens after the end of the project if nobody maintains the L3?	Content governance with named clinical responsibility (11); deliberately minimal technical infrastructure

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12.4 Regulatory Classification

> Lecture Day 5: *“Most CDSS tools fit in the definition of SaMD."*

Is WundCheck a medical device?

Yes. According to the IMDRF definition, *Software as a Medical Device* (SaMD) is software that is intended for one or more medical purposes and fulfils these purposes without being part of a hardware medical device. WundCheck serves the **detection and monitoring** of a disease (postoperative wound infection) and therefore falls under the definition.

The lecture names exceptions — national laws that exclude certain low-risk CDSS, a missing SaMD regulation, or a lack of enforcement. None of these plausibly applies to a patient-facing triage tool with emergency outputs.

IMDRF risk categorisation

	Inform clinical management	Drive clinical management	Treat or diagnose
Critical	IV	III	II
Serious	III	**II** ←	I
Non-serious	II	I	I

Our classification: Category II.

Rationale — the reasoning counts for more than the result:

- **State of healthcare situation = serious.** A postoperative wound infection is not self-limiting without treatment and can lead to sepsis, revision surgery or implant loss. It is not “critical" in the sense of an immediately life-threatening situation, but clearly more than “non-serious".
- **Significance of information = drive clinical management.** WundCheck does not make a diagnosis and does not propose a therapy — it triggers a contact and thereby directly influences *whether and when* clinical action is taken. That is more than “inform" (mere information), but less than “treat or diagnose".

Discussion point for the Q&A: One could argue for “critical", because a delayed infection carries the risk of sepsis — that would yield Category III. We decided against this because the IMDRF category describes the *typical* condition, not the rare extreme course. What matters is this: the lecture makes clear that regulators assess **each function individually** and that a single high-risk function raises the entire product into the stricter class.

Relevant standards

Standard	Subject	Relation to our project
ISO 14971	Risk management for medical devices	Our hazard analysis (10) is a simplified precursor
IEC 62304	Software life cycle	Our versioning and the change process (08) are first building blocks
IEC 62366	Usability engineering	The untested area — residual risk (b)
ISO 13485	Quality management system	Not addressed
GDPR / CH-DSG	Data protection	Health data are a special category; consent, purpose limitation and a deletion concept are required
EU AI Act	—	**Not applicable**: WundCheck is rule-based, not an AI system within the meaning of the regulation. A deliberate consequence of forgoing image analysis

What would be required for real-world deployment

- Conformity assessment and CE marking under MDR (CH: MepV)
- Quality management system according to ISO 13485
- Risk management file according to ISO 14971
- Clinical evaluation
- Usability engineering file according to IEC 62366
- Post-market surveillance

For our prototype the following applies: a disclaimer in the L3 (`meta.disclaimer`) and in the app — *"Study prototype FHNW MSc MI — not for clinical use."*

12.5 What We Would Do Differently Given More Time

Priority	Measure	Why
1	Clinical sign-off of all thresholds and rules	The single point that safeguards all eight L1→L2 decisions at once
2	Usability study on image assignment with laypersons	Validates the central design idea — currently the largest residual risk
3	Graphics for different skin types and body regions	Erythema looks different on dark skin; the current set depicts only one case
4	Conversion of the L3 to FHIR Questionnaire	Would establish the software neutrality that we deliberately forwent (08)
5	Multilingualism	Provided for in the L3 schema, but only `de` is populated — essential for the target group with language barriers
6	Offline queue with secured synchronisation	Currently only a manual FHIR download

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